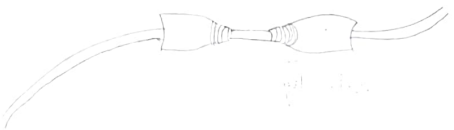
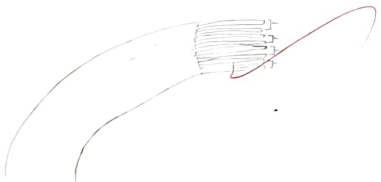
Step 1: To start construction of the device, begin by threading shields onto the cable.



Step 2: Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.



Step 3: After, you will need to untangle the wires; there should be 4 twisted pairs. Referencing back to the slot arrange from top to bottom. One end should be in arrangement A and the other in B.



Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you.



Ex. 02

DATE: 23/07/2026

# Network Cables

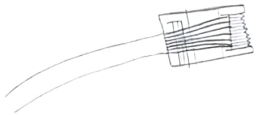
AIM: Study of different types of Network Cables.

(a) Understand different types of network cable.

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

| CABLE TYPE | CATEGORY | MAXIMUM DATA TRANSMISSION | ADVANTAGES/ DISADVANTAGES | APPLICATION / USE |
|---|---|---|---|---|
| UTP | Category 3 | 10 bps | Advantages:<br>* cheaper in cost.<br>* Easy to install as they have a small overall diameter.<br>Disadvantages:<br>* More prone to EMI (Electromagnetic interference & noise | 1. 10 Base - T Ethernet<br>2. Fast Ethernet, Gigabit Ethernet.<br>3. Fast Ethernet, Gigabit Ethernet. |
| | Category 5 | upto 100 Mbps | | |
| | Category 5e | 1 GbPS | | |
| STP | Category 6, 6a | 10 Gbps | Advantages:<br>* shielded<br>* Faster than UTP<br>* Less susceptible to noise & interference.<br>Disadvantage:<br>* Expensive<br>* Greater<br>* installation effort | 1. Gigabit Ethernet, 10G Ethernet (55 m) - widely used in data centres<br>2. Gigabit Ethernet, 10G ethernet (100 m) |
| SSTP | Category 7 | | | |
| Coaxial Cable | RG - 6<br>RG - 59<br>RG - 11 | 10 - 100 Mbps | Advantages:<br>* High bandwidth<br>* Immune to interference<br>* Low loss band width<br>* Versatile.<br>Disadvantage:<br>* Cost<br>* Size is bulky | Speed of signal is 500m Television network High speed internet connection. |

Step5: Next, push the cable right in. The notch at the end the plug needs to be just over the cable shielding. and if it isn't, that means that you stripped off too much shield. Simply strip the cables back a little more.



Step6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step7: Lastly, repeat for the other end using diagram B (to make Crossover cables) / using diagram A (to make a straight through cable).

| Cable Type | Category | Maximum data Transfer | Advantages / Disadvantages. | Application / use. |
|---|---|---|---|---|
| Fibre optics Cable. | Single mode<br>Multimode | 100 Gbps | Advantages:<br>★ High speed<br>★ High bandwidth<br>★ High security<br>★ Long distance.<br>Disadvantages:<br>★ Expensive<br>★ Required skilled installers | Maximum distance of fibre optics cable is around 100 meters. |

(b) Make Your own Ethernet Cross-Over / straight Cable.

Tools and parts needed.

★ Ethernet Cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.

★ A crimping Tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.

★ Two RJ45 plugs.

★ Optional 2 plug shields.

RESULT:

Thus the different types of Eab network cables are studied successfully.